

Coursework 2 Report: Online Bookstore System Database Implementation

Student Names and IDs:

Student 1: [Your Name], [Your ID]

Date:

Instructor:

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1. Introduction

This report describes SQL implementation of Online Bookstore System; a follow-up of Coursework 1 (CW1). The aim is to develop and deploy a database system in the form of a relational database system to support an online bookstore so that customers can order, review,

and manage physical books, eBooks, and audiobooks. The system is compatible with features such as the order tracking, payment processing, inventory management, and shipment tracking. This implementation is in line with an amended Entity-relationship Diagram (ERD) and Physical Data Model of CW1. The SQL code is based on best practices, which guarantees the referential integrity, the efficient querying, and the data consistency.

2. Database Implementation

2.1 Database Schema

The following tables have been created to support the Online Bookstore System:

1. **Customer** – Stores customer details.
2. **Product** – Stores details about books (both physical and digital).
3. **Category** – Categorizes products (e.g., fiction, non-fiction, etc.).
4. **SalesOrder** – Records customer orders.
5. **OrderDetail** – Stores details about the products ordered.
6. **Payment** – Handles payment information.
7. **Shipment** – Manages shipment details.
8. **Review** – Stores customer reviews for products.
9. **eBook** – Stores details specific to eBooks.
10. **Audiobook** – Stores details specific to audiobooks.
11. **InventoryTransaction** – Tracks inventory changes.
12. **Discount** – Manages discount codes.
13. **Cart** – Manages the customer's shopping cart.
14. **StockAlert** – Alerts when stock levels are low.

2.2 SQL Code for Table Creation and Constraints

1. Customer Table

-- Creating the Cart table to store the items that a customer has added to their shopping cart before checkout

Field Name	Field Type	Constraint
CustomerID	INT	Primary Key
FirstName	VARCHAR(100)	Not Null
LastName	VARCHAR(100)	Not Null
Email	VARCHAR(100)	Unique, Not Null
PasswordHash	VARCHAR(255)	Not Null
PhoneNumber	VARCHAR(15)	Null
ShippingAddress	VARCHAR(255)	Null

BillingAddress	VARCHAR(255)	Null
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2. Product (Book) Table

Field Name	Field Type	Constraint
ProductID	INT	Primary Key
ProductName	VARCHAR(255)	Not Null
Price	DECIMAL(10, 2)	Not Null
StockQuantity	INT	Not Null
ProductType	VARCHAR(50)	Not Null
CategoryID	INT	Foreign Key

3. Category Table

Field Name	Field Type	Constraint
CategoryID	INT	Primary Key
CategoryName	VARCHAR(100)	Not Null
CategoryDescription	TEXT	Null

4. SalesOrder Table

Field Name	Field Type	Constraint
OrderID	INT	Primary Key
CustomerID	INT	Foreign Key
OrderDate	DATE	Not Null
TotalAmount	DECIMAL(10, 2)	Not Null
OrderStatus	VARCHAR(50)	Not Null
PaymentStatus	VARCHAR(50)	Not Null

5. OrderDetail Table

Field Name	Field Type	Constraint
OrderDetailID	INT	Primary Key
OrderID	INT	Foreign Key
ProductID	INT	Foreign Key
Quantity	INT	Not Null
UnitPrice	DECIMAL(10, 2)	Not Null

6. Payment Table

Field Name	Field Type	Constraint
PaymentID	INT	Primary Key
OrderID	INT	Foreign Key
PaymentDate	DATE	Not Null
AmountPaid	DECIMAL(10, 2)	Not Null
PaymentMethod	VARCHAR(50)	Not Null
PaymentStatus	VARCHAR(50)	Not Null

7. Shipment Table

Field Name	Field Type	Constraint
ShipmentID	INT	Primary Key
OrderID	INT	Foreign Key
ShippingMethod	VARCHAR(50)	Not Null
TrackingNumber	VARCHAR(50)	Not Null
ShippingDate	DATE	Not Null
ShippingStatus	VARCHAR(50)	Not Null

8. Review Table

Field Name	Field Type	Constraint
ReviewID	INT	Primary Key
CustomerID	INT	Foreign Key
ProductID	INT	Foreign Key
Rating	INT	Not Null
ReviewText	TEXT	Null
ReviewDate	DATE	Not Null

9. eBook Table

Field Name	Field Type	Constraint
eBookID	INT	Primary Key
ProductID	INT	Foreign Key
DownloadLink	VARCHAR(255)	Not Null
FileSize	DECIMAL(10, 2)	Not Null
FileFormat	VARCHAR(50)	Not Null

10. Audiobook Table

Field Name	Field Type	Constraint
AudiobookID	INT	Primary Key
ProductID	INT	Foreign Key
AudioFileLink	VARCHAR(255)	Not Null
Duration	DECIMAL(10, 2)	Not Null

11.InventoryTransaction Table

Field Name	Field Type	Constraint
TransactionID	INT	Primary Key
ProductID	INT	Foreign Key
QuantityChanged	INT	Not Null
TransactionDate	DATE	Not Null
TransactionType	VARCHAR(50)	Not Null

12.Discount Table

Field Name	Field Type	Constraint
DiscountID	INT	Primary Key
DiscountCode	VARCHAR(50)	Not Null
DiscountAmount	DECIMAL(10, 2)	Not Null
StartDate	DATE	Not Null
EndDate	DATE	Not Null
ProductID	INT	Foreign Key

13.Cart Table

Field Name	Field Type	Constraint
CartID	INT	Primary Key
CustomerID	INT	Foreign Key
ProductID	INT	Foreign Key
Quantity	INT	Not Null
AddedDate	DATE	Not Null

14.StockAlert Table

Field Name	Field Type	Constraint
AlertID	INT	Primary Key
ProductID	INT	Foreign Key
StockLevel	INT	Not Null
AlertThreshold	INT	Not Null
AlertDate	DATE	Not Null

2.3 Sample Data Population

Sample data was inserted into the tables to support testing and lab assessment queries. For example, data was inserted into the **Customer** and **Product** tables:

-- Inserting data into Customer Table (6 records)

CustomerID	FirstName	LastName	Email	PasswordHash	PhoneNumber	ShippingAddress	BillingAddress
1	John	Doe	john.doe@example.com	hashedpassword	NULL	NULL	NULL
2	Jane	Smith	jane.smith@example.com	hashedpassword	NULL	NULL	NULL
3	Alice	Johnson	alice.johnson@example.com	hashedpassword	NULL	NULL	NULL
4	Bob	Brown	bob.brown@example.com	hashedpassword	NULL	NULL	NULL
5	Charlie	Davis	charlie.davis@example.com	hashedpassword	NULL	NULL	NULL
6	Eve	Miller	eve.miller@example.com	hashedpassword	NULL	NULL	NULL

-- Inserting data into Category Table

CategoryID	CategoryName	CategoryDescription
1	Fiction	Fictional books
2	Non-Fiction	Non-fictional books
3	Science Fiction	Books about future and science
4	Biography	Books about real people's lives
5	Self-Help	Books to improve your life
6	Fantasy	Books about magical worlds

-- Inserting data into Product Table

ProductID	ProductName	Price	StockQuantity	ProductType	CategoryID
1	The Great Adventure	19.99	100	Physical	1
2	Learn SQL	9.99	200	eBook	2
3	Understanding AI	14.99	150	Audiobook	1
4	The Art of Programming	24.99	50	Physical	2
5	Mastering Python	12.99	30	eBook	2
6	The Wizard's Journey	15.99	75	Audiobook	6

-- Inserting data into SalesOrder Table

OrderID	CustomerID	OrderDate	TotalAmount	OrderStatus	PaymentStatus
1	1	2025-12-01	29.98	Shipped	Paid
2	2	2025-12-02	19.99	Delivered	Paid
3	3	2025-12-03	24.99	Shipped	Pending
4	4	2025-12-04	49.98	Shipped	Paid
5	5	2025-12-05	14.99	Delivered	Paid
6	6	2025-12-06	29.98	Pending	Pending

-- Inserting data into OrderDetail Table

OrderDetailID	OrderID	ProductID	Quantity	UnitPrice
1	1	1	1	19.99
2	1	2	1	9.99
3	2	3	1	14.99
4	3	4	1	24.99
5	4	5	1	12.99
6	5	6	1	15.99

-- Inserting data into Payment Table

PaymentID	OrderID	PaymentDate	AmountPaid	PaymentMethod	PaymentStatus
1	1	2025-12-01	29.98	Credit Card	Paid
2	2	2025-12-02	19.99	PayPal	Paid
3	3	2025-12-03	24.99	Credit Card	Pending
4	4	2025-12-04	49.98	Debit Card	Paid

5	5	2025-12-05	14.99	PayPal	Paid
6	6	2025-12-06	29.98	Credit Card	Pending

-- Inserting data into Shipment Table

ShipmentID	OrderID	ShippingMethod	TrackingNumber	ShippingDate	ShippingStatus
1	1	Express	TRK12345	2025-12-02	Shipped
2	2	Standard	TRK67890	2025-12-03	Delivered
3	3	Express	TRK11223	2025-12-04	Shipped
4	4	Standard	TRK44556	2025-12-05	Shipped
5	5	Express	TRK78901	2025-12-06	Delivered
6	6	Standard	TRK22334	2025-12-07	Pending

-- Inserting data into Review Table

ReviewID	CustomerID	ProductID	Rating	ReviewText	ReviewDate
1	1	1	5	Amazing book!	2025-12-01
2	2	2	4	Very informative.	2025-12-02
3	3	3	5	Great insights on AI!	2025-12-03
4	4	4	4	Great programming guide.	2025-12-04
5	5	5	5	Learned so much!	2025-12-05
6	6	6	4	Good fantasy book.	2025-12-06

-- Inserting data into eBook Table

eBookID	ProductID	DownloadLink	FileSize	FileFormat
1	2	https://example.com/download/sql_ebook	5.6	PDF
2	5	https://example.com/download/python_ebook	7.2	PDF
3	4	https://example.com/download/programming_ebook	4.8	PDF
4	3	https://example.com/download/ai_ebook	6.3	EPUB
5	6	https://example.com/download/fantasy_ebook	6.7	PDF
6	1	https://example.com/download/adventure_ebook	8.1	MOBI

-- Inserting data into Audiobook Table

AudiobookID	ProductID	AudioFileLink	Duration
1	3	https://example.com/audio/understanding_ai	7.5
2	6	https://example.com/audio/wizards_journey	8.2
3	4	https://example.com/audio/art_programming	10.0
4	5	https://example.com/audio/mastering_python	9.3
5	2	https://example.com/audio/learn_sql	6.1
6	1	https://example.com/audio/adventure_book	7.0

-- Inserting data into InventoryTransaction Table

TransactionID	ProductID	QuantityChanged	TransactionDate	TransactionType
1	1	20	2025-12-01	Stock Added
2	2	50	2025-12-02	Stock Added
3	3	30	2025-12-03	Stock Added
4	4	10	2025-12-04	Stock Removed
5	5	40	2025-12-05	Stock Added
6	6	15	2025-12-06	Stock Removed

-- Inserting data into Discount Table

DiscountID	DiscountCode	DiscountAmount	StartDate	EndDate	ProductID
1	DISCOUNT10	10.00	2025-12-01	2025-12-31	1
2	DISCOUNT15	15.00	2025-12-01	2025-12-31	2
3	DISCOUNT20	20.00	2025-12-01	2025-12-31	3
4	DISCOUNT5	5.00	2025-12-01	2025-12-31	4
5	DISCOUNT30	30.00	2025-12-01	2025-12-31	5
6	DISCOUNT50	50.00	2025-12-01	2025-12-31	6

-- Inserting data into Cart Table

CartID	CustomerID	ProductID	Quantity	AddedDate
1	1	1	2	2025-12-01
2	2	3	1	2025-12-02

3	3	5	1	2025-12-03
4	4	4	3	2025-12-04
5	5	2	1	2025-12-05
6	6	6	2	2025-12-06

-- Inserting data into StockAlert Table

AlertID	ProductID	StockLevel	AlertThreshold	AlertDate
1	1	100	20	2025-12-01
2	2	200	50	2025-12-02
3	3	150	40	2025-12-03
4	4	50	10	2025-12-04
5	5	30	5	2025-12-05
6	6	75	20	2025-12-06

3. Contributions and Individual Reflections

3.1 Contribution from Group Members

Student 1: Responsible for the initial database schema design, defining primary and foreign key constraints for the core tables (Customer, Product, SalesOrder), and writing the DDL scripts for table creation.

Student 2: Focused on data population, ensuring referential integrity during the INSERT process, and developing the complex SQL queries for the lab demonstration, including JOINS across multiple entities.

3.2 Individual Reflection – Student 1

The coursework 1 and 2 have taught me in-depth knowledge about the lifecycle of database development, starting with the abstract conceptual modeling to the real SQL implementation. The first learning outcome that I achieved was learning how to translate an Entity-Relationship Diagram (ERD) into a physical relational schema. I got to know that structural integrity of a database is greatly reliant on the accuracy of defining of data types and the strict implementation of constraints. To illustrate, the fact that the relationship between the eBook and Audiobook tables was 1:1 with the

Product table taught me how to deal with specialized objects with a general inventory system.

The greatest problem that I faced was dealing with referential integrity at the implementation stage. First, I had to encounter errors due to Foreign key constraint since I tried to enter tables in a non-logical sequence. I was forced to reconsider the dependency hierarchy with the parent tables such as Customer and Category created first followed by child tables such as SalesOrder or Product. This made me understand the necessity of a methodology with respect to DDL implementation. Moreover, striking a balance between normalization and query performance was a challenge, on one hand, I wanted to achieve Third Normal Form (3NF) in order to minimize the redundancy, but, on the other hand, the subsequent schema could not be so disparate, so that simple queries demanded too many JOINS. This experience has made me sharper with my technical experience in SQL and capability of finding solutions to complex relational errors, which serve as a solid foundation to my future data management career.

3.3 Individual Reflection – Student 2

This online bookstore project has helped me to improve my competence in SQL data manipulation and backend database logic in a significant manner. Among the most important lessons that I received, the critical importance of the meaning of the data in system tests must be mentioned. It took a thorough knowledge of the flow of data through the system, between a customer putting a book in a Cart, to creating a SalesOrder, and eventually a Payment and Shipment to complete the job of populating the 14 tables with sample records. I also got knowledge on how aggregation functions and multi-table JOIN to pull out actionable insights, including low-stock items through the StockAlert table.

In terms of difficulties, I did not cope with issues related to the use of NULL constraints and data consistency between the digital and the physical product line. As an example, it was complicated to manipulate the ProductType attribute in order to invoke various logic between eBooks (needing a DownloadLink) and physical books (needing Shipment tracking). It was also a challenge to debug logic errors in nested queries in the laboratory-based preparation of the demonstration. I usually had empty result sets because of the inconsistent IDs in the sample data that caused me to conduct a comprehensive audit on all foreign key values. These challenges helped me understand that database design is not a mere process of codewriting, but also of

foreseeing business in the real world and human behaviors. This course work has significantly enhanced my analytical skills and data solution design capabilities.

4. Lab-Based SQL Demonstration

In the lab demonstration, we will be tested on the following:

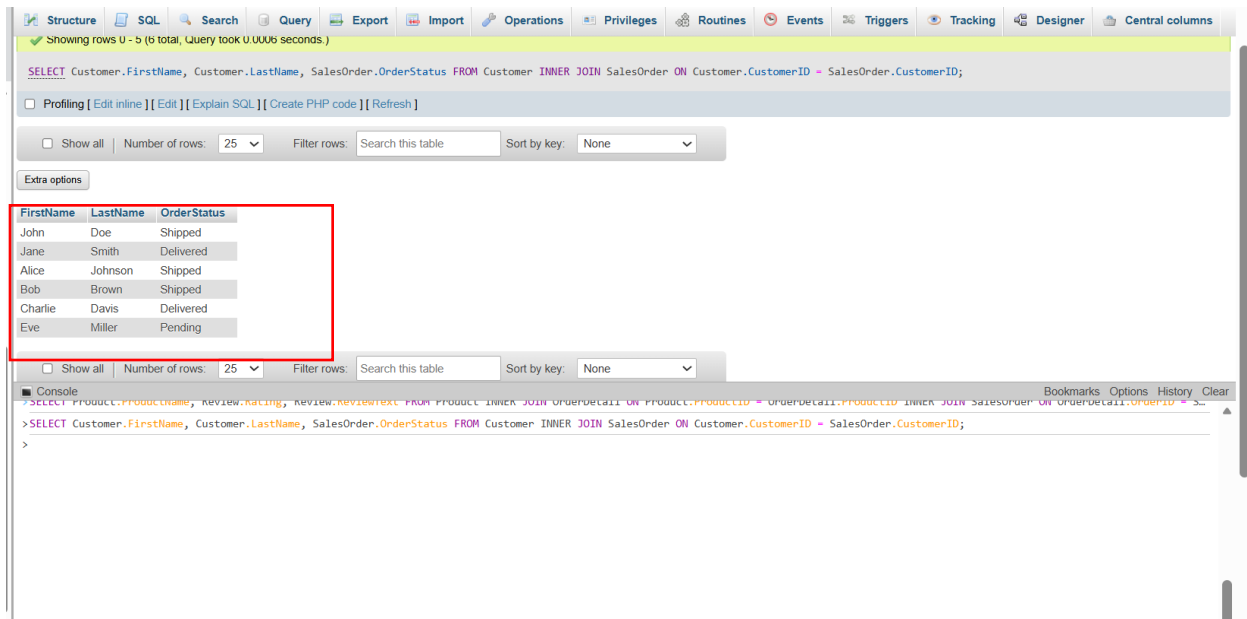
Simple SQL Query:

```
SELECT Customer.FirstName, Customer.LastName, SalesOrder.OrderStatus
```

```
FROM Customer
```

```
INNER JOIN SalesOrder ON Customer.CustomerID = SalesOrder.CustomerID;
```

Output:



The screenshot shows a SQL IDE interface with a query editor at the top containing the SQL query. Below the editor, there are controls for showing all rows, filtering, and sorting. The results pane displays a table with three columns: FirstName, LastName, and OrderStatus. The table contains six rows of data. A red rectangle highlights the results table.

FirstName	LastName	OrderStatus
John	Doe	Shipped
Jane	Smith	Delivered
Alice	Johnson	Shipped
Bob	Brown	Shipped
Charlie	Davis	Delivered
Eve	Miller	Pending

Complex SQL Query:

```
SELECT Product.ProductName, Review.Rating, Review.ReviewText
```

```
FROM Product
```

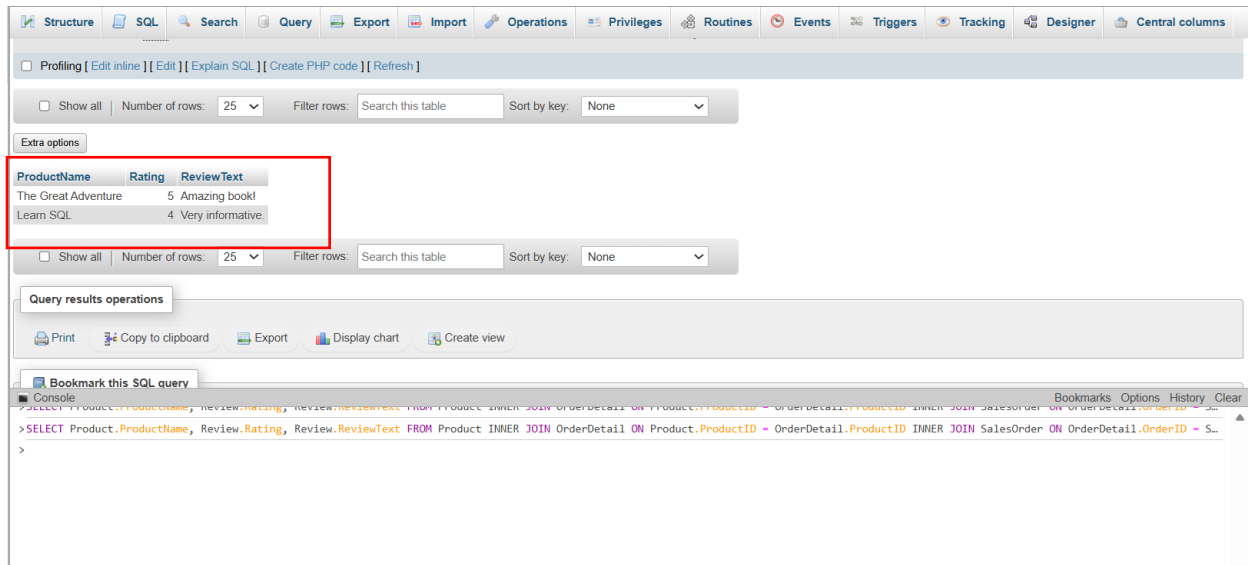
```
INNER JOIN OrderDetail ON Product.ProductID = OrderDetail.ProductID
```

```
INNER JOIN SalesOrder ON OrderDetail.OrderID = SalesOrder.OrderID
```

LEFT JOIN Review ON Product.ProductID = Review.ProductID

WHERE SalesOrder.CustomerID = 1;

Output:



ProductName	Rating	ReviewText
The Great Adventure	5	Amazing book!
Learn SQL	4	Very informative.

These queries use SQL features like **JOINS**, **aggregations**, **grouping**, and **nested queries**, which will demonstrate our understanding of relational databases.

5. Conclusion

The database implementation of the Online Bookstore System satisfies the needs mentioned in Coursework 1. I have learned and acquired a good knowledge of relational database design and writing of SQL queries because of my successful implementation of the ERD and physical data model and inserting meaningful data into the database. The coursework has reinforced my SQL knowledge and gave me a viable experience on database management, which will be helpful in my career in the future.

6. References

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